

Element C student work

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For this project to be successful it is clear that the water conditions and ability of the turtles to have access to basking is essential.

Primary Stakeholders/Water Conditions

The primary stakeholders for this project are the Turtles: Bella and Zork, and their caregivers. It is important for Bella and Zork's habitat to receive maintenance to ensure their needs and water conditions are at levels that support their health and wellbeing. At the current moment to maintain necessary conditions in tank, there must be a filtration process that takes place. The tank must be fully drains every three to four weeks, taking the caretakers about 2 hours to complete the process.

Primary Stakeholders/ Basking Conditions

The primary stakeholders remain the same. Turtles are not warm blooded they need to warm up through basking to regulate their body temperature and ensure shell health by receiving IR and UV light from the sun, or in our situation a lamp. Our current problem is there is not enough room for both turtles to bask at the simultaneously, typically placing the male turtle Zork at a disadvantage.

List of Requirements/Quantity of Design Requirements

1. Maintaining proper water conditions for the turtles
2. Providing a place for Bella and Zork to receive UV/IR that includes space for both turtles
3. Efficient Filtration
4. Reduce Maintenance Time for Caretakers
5. Expansion of Tank- Recreation and Functional

Proposed Solutions/Primary Stakeholders

1. *Maintaining proper water conditions for the turtles*
 - This solution ensures that all requirement to ensure the health and wellbeing of the turtles are maintained
2. *Providing a place for Bella and Zork to receive UV/IR (FOCUS)*
 - This solution is to make sure that both turtles receive proper UV/IR to ensure shell health and to assist with the turtles warming up. This solution also addresses the previous basking area of the tank by creating a spot for both turtles to receive light, as to the previous system which allowed only one turtle at a time.

3. *Reduce Maintenance Time for Caretakers*

- Every 3-4 weeks for hours the caretakers of the tank are spending time resetting the task to create a sustainable environment for the turtles. 2 Hours seems like a lot of time, so including a way to make the process more efficient or maintaining the current time is essential. The goal is to simplify the process not make it harder or more time consuming if possible.

4. *Expansion of Tank- Recreation and Functional*

- Bella is much larger than Zork and I have heard from classmates that this sometimes causes conflict usually with Zork being the party that receives the disadvantages. Creating a safe space for Zork and Bella that allows Zork to interact but also retreat is a consideration that I believe should be considered to promote the wellbeing of the turtles and to prohibit stress levels for Zork.

Conclusion

In conclusion our group plans to improve upon current conditions by improving the maintenance process for the caretakers(primary) and providing revisions to the basking platform for the turtles (primary). Our first goal however is to maintain the health and wellbeing of the turtles in their habitat aiming it improve processes not make them more difficult for the stakeholders. Our proposed requirements and solutions are listed above.